

Information for First and Second Responders

Vehicle Emergency Response Guide (ERG)

ELECTRIC  PROPULSION	VEHICLE NAME/MODEL: GILLIG 35/40-foot Hybrid Electric Bus  GUIDE FILENAME: GILLIG-ALLISONeGenFlex-HEV-35/40.pdf VEHICLE TYPE/DESIGNATION: Heavy Duty Class-8 Urban Bus	Type of RESS: Lithium-Ion, 644V battery storage system
		
Version 2022-10-21 PN 51-84891-003		

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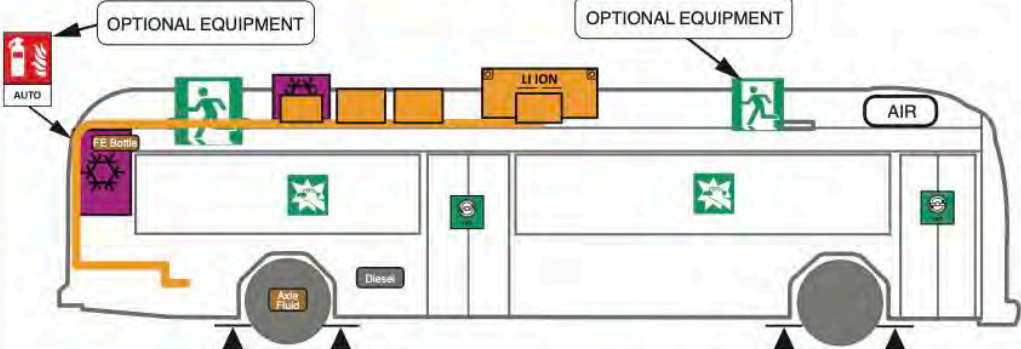
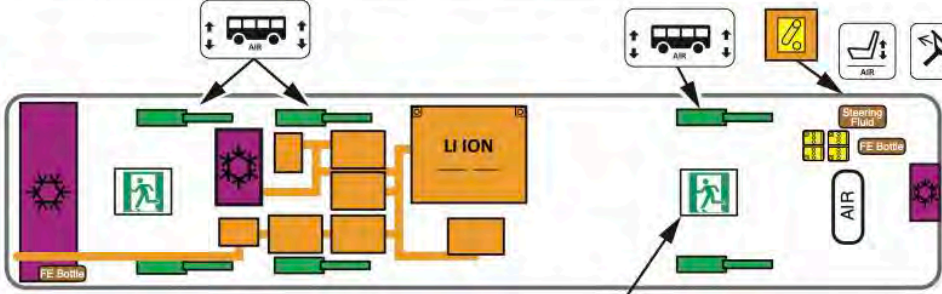
0. Rescue sheet



VEHICLE NAME/MODEL: GILLIG 35/40-foot Hybrid Electric Bus
VEHICLE TYPE/DESIGNATION: Heavy Duty Class-8 Urban Bus
PRODUCTION YEARS: 2021 –

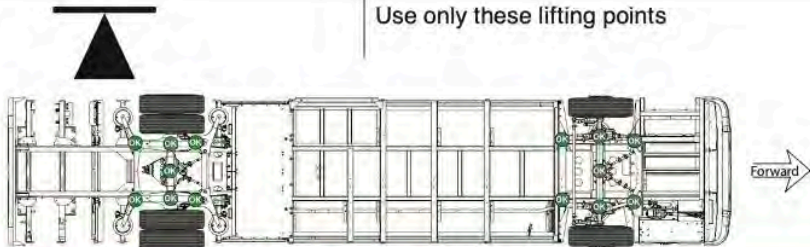




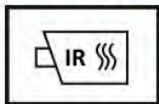


					
Electric propulsion		High-voltage Li-ion battery	High-voltage component	High-voltage power cable	Disconnect high-voltage
					
Low-voltage battery	Emergency exit	Break to obtain access	Emergency door	Auto fire suppression	Air conditioning component
					
Suspension Air Spring	Lifting point	Height control	Seat adjustment	Air tank	Steering Wheel Control
			ID No.	Version No.	Page No.

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1. Identification / Recognition	
	Hybrid Electric bus
2. Immobilisation / Stabilisation / Lifting	
	Bus height control in driver cabin
	Steering wheel, tilt control
	Seat height adjustment in driver cabin
	Use only these lifting points
3. Disable direct hazards / Safety regulations	
	Shutdown high voltage When circumstances allow, park the bus in a safe location, set the parking brake, and select "neutral" on the shift selector. Use 12/24V switch to disable HV 
4. Access to the occupants	
	One front door exit Rear door is optional
	One rear roof hatch standard An additional roof hatch is optional
	Break these windows to obtain access

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5. Stored Energy / Liquids / Solids			
			
			
			
			
			
6. In case of fire			
			
			
7. In case of submersion			
		Follow heading 3. Disable direct hazards / safety regulations once out of the water	
8. Towing / Transportation / Storage			
		Check battery temperature Remove driveshaft/axle shafts for towing	

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1. Identification / recognition

This chapter provides relevant information on identification of a GILLIG Hybrid Electric bus.

1.1 Special badging

Badging is used to identify GILLIG hybrid electric vehicles, as shown in Fig. 1.1.

These emblems are applied in the vehicle front, rear, street, and curbside locations, as shown in Fig. 1.2.

Emergency vehicle identification label is located under rear vehicle door, as shown in Fig. 1.3.

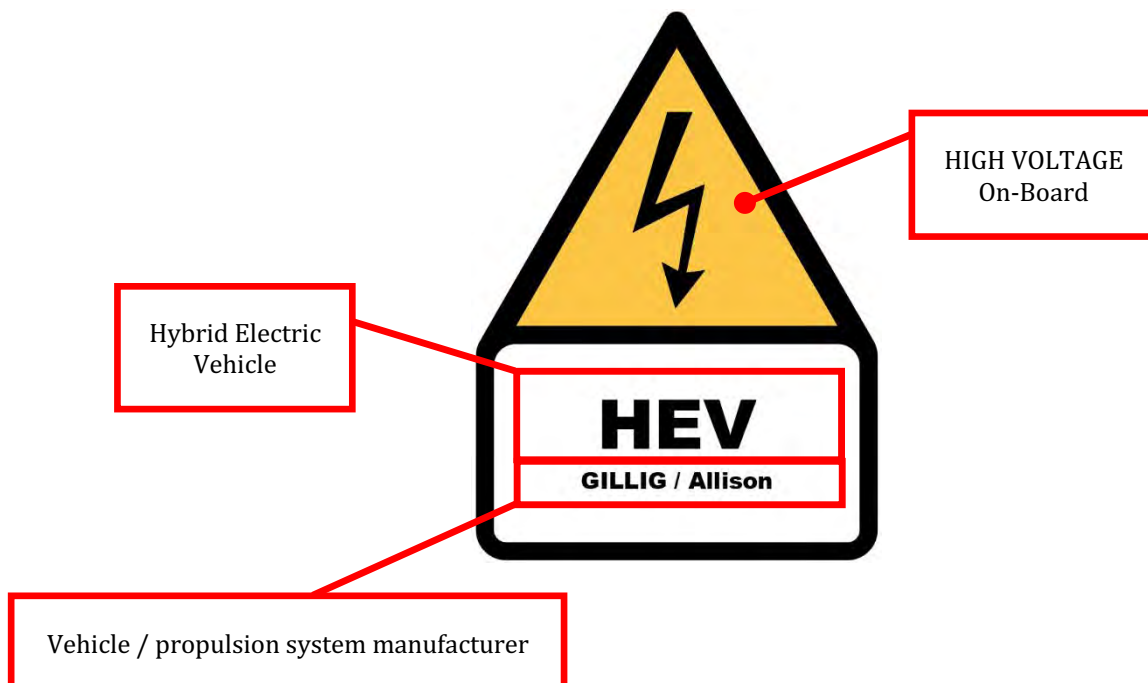


Fig. 1.1. GILLIG Hybrid Electric Vehicle Badging

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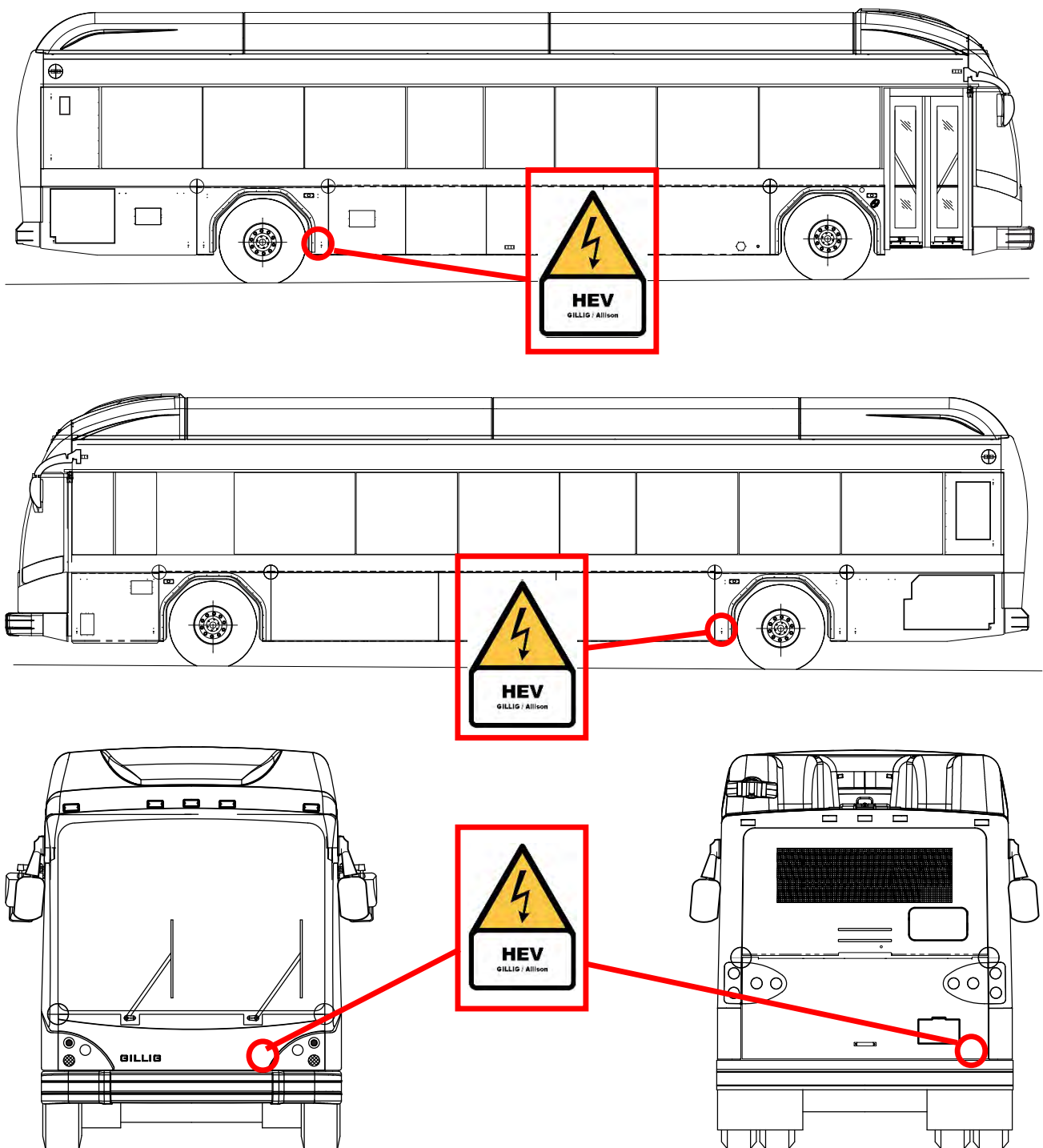


Fig. 1.2. GILLIG Hybrid Electric Vehicle Badging Locations

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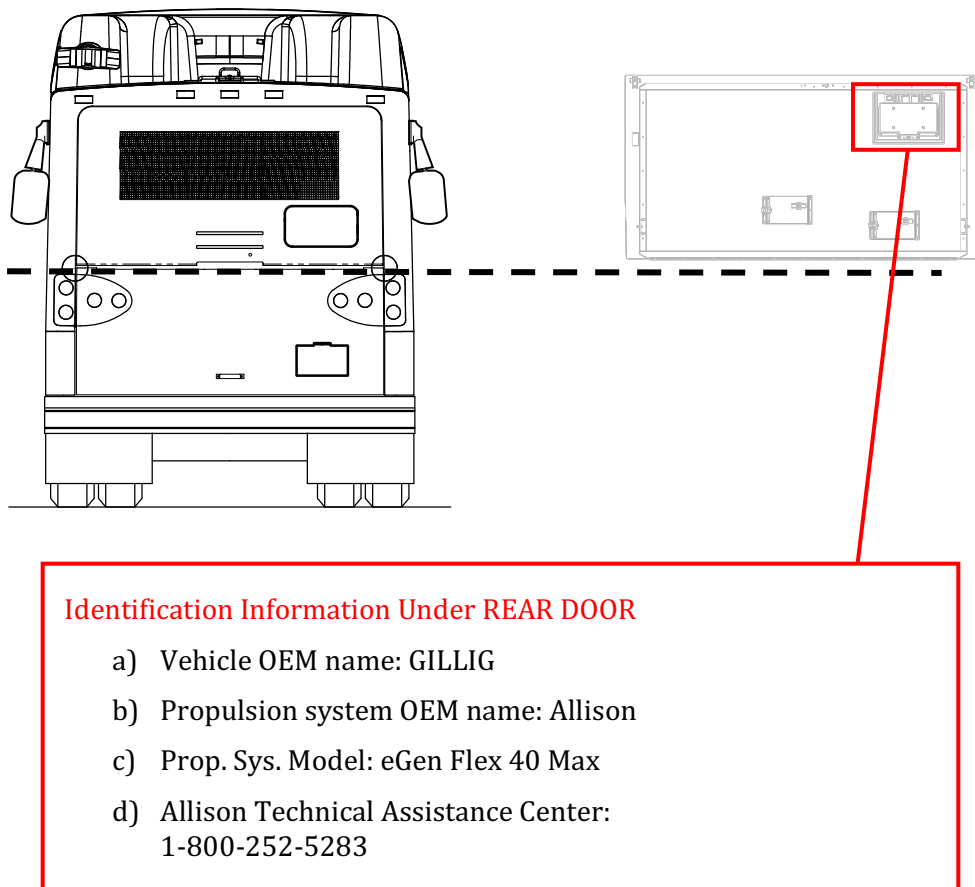


Fig. 1.3. GILLIG Emergency Vehicle Identification Label

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1.2 Vehicle identification plate

The VIN plate contains information for the identification of the vehicle model (35 or 40 foot) and hybrid electric propulsion, as shown in Fig. 1.4.

15GGB301_K3serialnumber

35-foot Hybrid

15GGD301_K3serialnumbr

40-foot Hybrid

Fig. 1.4. GILLIG VIN Denomination for 35/40-foot Hybrid Electric Bus

The location of the VIN plate is shown in Fig. 1.5 in the upper right corner above the driver's seat.

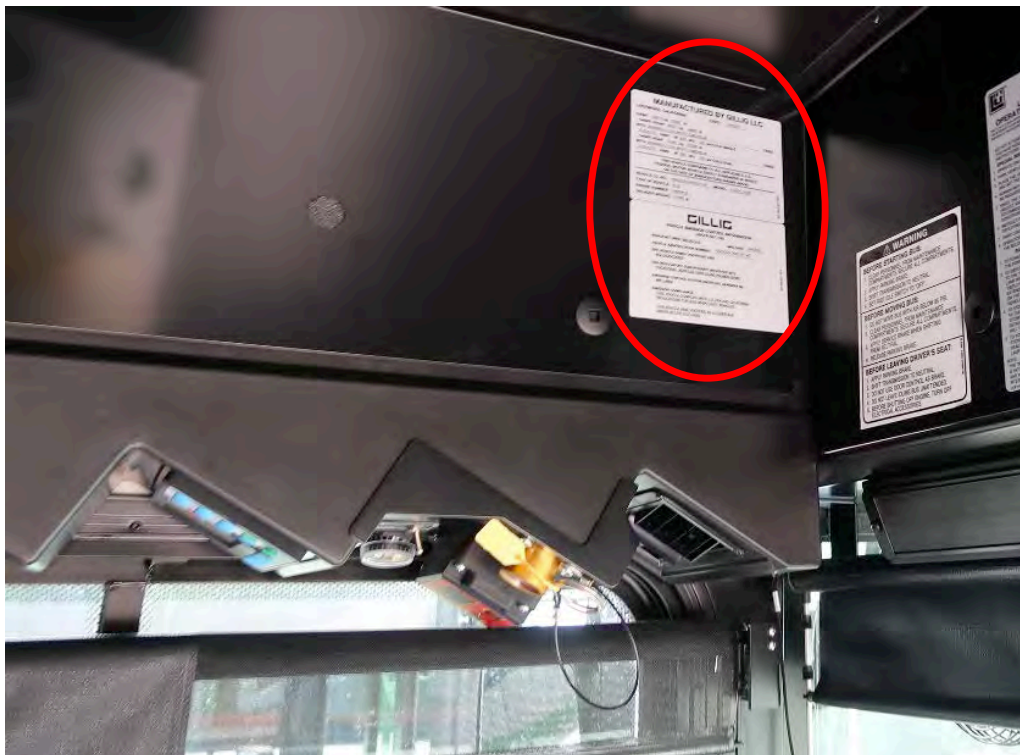


Fig. 1.5. GILLIG VIN Plate Location

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1.3 HEV label

There is no interior label for HEV.

1.4 Information on High-Voltage Energy Source

The Lithium-Ion battery cell chemistry supplies the hybrid system components with 644 VDC nominal voltage and can range from 450-728 VDC. The bus is equipped with one battery pack with a useable capacity of 20 kWh. The High Voltage system layout is shown in Fig. 1.6.

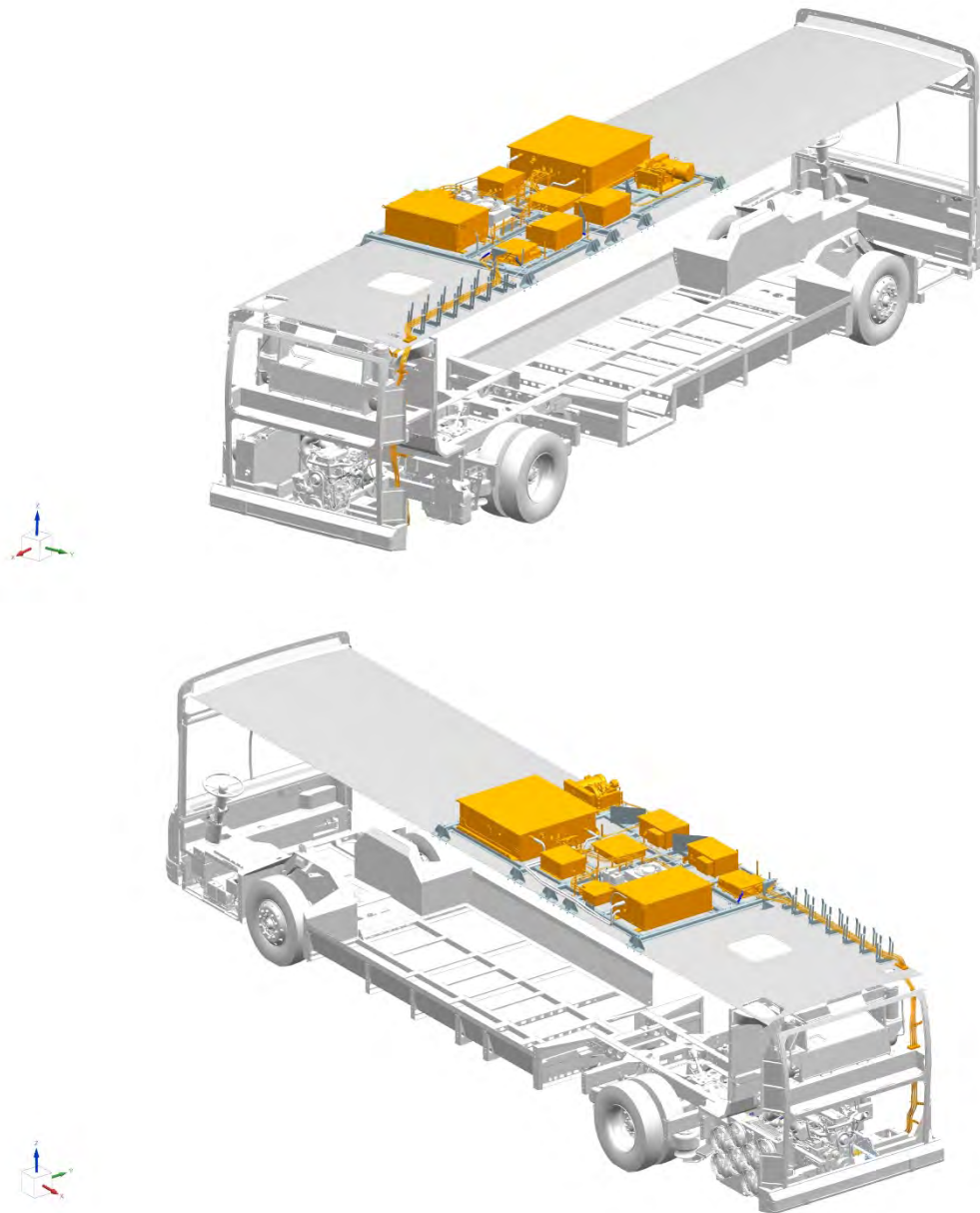


Fig. 1.6. GILLIG Hybrid Electric Bus High Voltage System Layout

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2. Immobilization / stabilization / lifting

To turn the vehicle OFF, follow either the A) “non-emergency” immobilization procedure or the B) “emergency” immobilization procedure, depending on the specific circumstances.

A) **Non-emergency** immobilization procedure:

1. Park the bus in a safe location;
2. Set the parking brake and select “Neutral” on the push-button Shift selector and turn the Master Run Control to the “OFF” position, as indicated in Fig. 2.1:

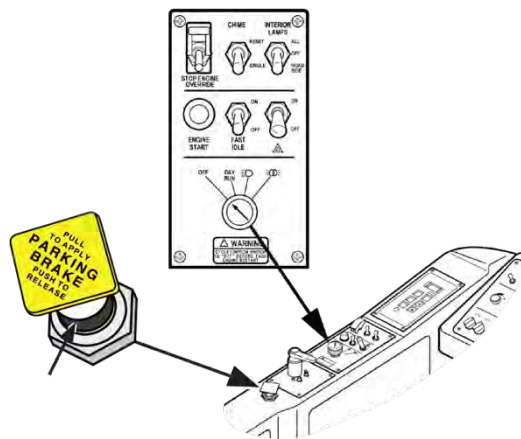


Fig. 2.1

3. Release the air pressure from the front door motors by moving the “Door Air” lever (located at the rear of the driver’s left console) to the “RELEASE” position. This action avoids automatic door closure;
4. Wait at least 60 seconds;
5. Disable the system by flipping the 12/24V front disconnect switch downward, as show in Fig. 2.2:

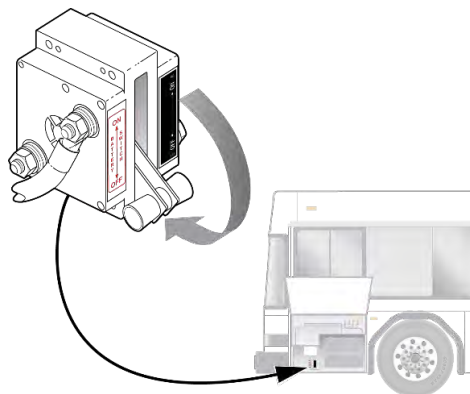


Fig. 2.2

6. Complete the electrical disconnect verification procedure (see Allison Service Manuals SM8685) prior to accessing high-voltage components.

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B) Emergency immobilization procedure:

1. Disable the system by flipping the 12/24V front disconnect switch downward, as show in Fig. 2.3:

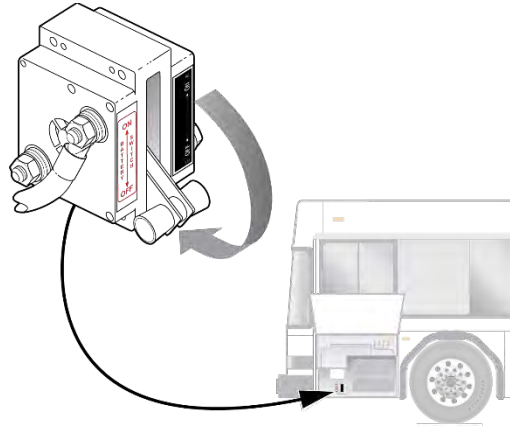


Fig. 2.3

2. If possible, set the parking brake and select “Neutral” on the push-button Shift selector.

C) If jacking is required, follow the procedure indicated here or in the vehicle operator’s manual:

1. Position the bus on a hard, level, flat surface before jacking.
2. Set the parking brake and select “Neutral” on the push-button Shift selector and turn the Master Run Control to the “OFF” position, as indicated in Fig. 2.4:

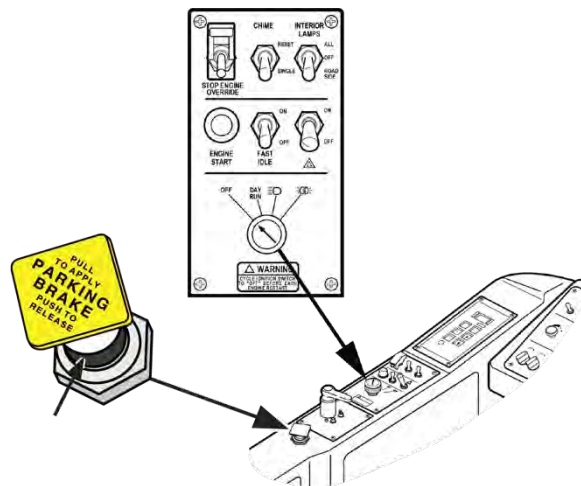


Fig. 2.4

3. Wait at least 60 seconds.
4. Turn the Main Disconnect Switch (12/24V) to the “OFF” position.
5. Install the LOTO lock and LOTO tag onto the Main Disconnect Switch (12/24V) to lock it in the “OFF” position.

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6. The floor jack must be located properly (Fig. 2.5) to prevent the possibility of the jack sliding from under the bus.

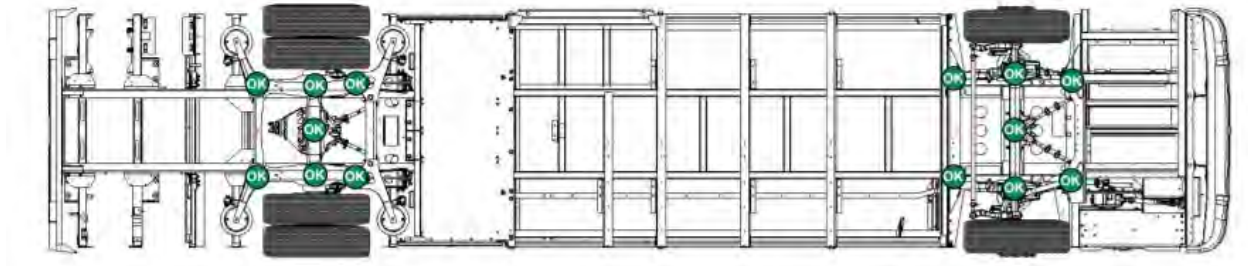


Fig. 2.5

Do not position the jack under the “dropped” frame rails in the center section of the bus!

7. Block or chock the front and rear of the wheel on the opposite side of the bus being raised to help prevent the bus from moving.
8. If a flat tire is to be removed, raise the bus so an inflated spare tire will just clear the surface. Jacking the bus to change a front tire can be done on the front axle beam, and at other frame locations indicated by the OK symbols in Figure 1-25. Jacking the bus to change a rear tire can be done on the rear axle and at the locations on the rear suspension indicated by an OK symbol.

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3. Disable direct hazards / safety regulations

Use the “IDENTIFY/IMMOBILIZE/DISABLE” emergency process, as referenced by National Fire Protection Agency (NFPA) “Emergency Field Guide” Ed. 2018:

IDENTIFY

- Look for external badging indicating the alternative fuel vehicle. See “identification” section.
- This vehicle has external high-voltage identification badging and warning labels per SAE J2990 and J2910.
- Beware, badging may be hidden in a crash or fire, so alternative identification methods may need to be used. Refer to the “identification” section for alternate identification methods.
- When circumstances allow, determine vehicle’s make, model, and year (see “identification” section).

IMMOBILIZE

- Vehicles should be immobilized prior to working around them
- Approach the vehicle from a 45° angle to stay out of the potential path of travel and chock the wheels.
- Refer to Chapter 2 “Immobilization” for “non-emergency” and “emergency” immobilization procedures specific for this vehicle.

DISABLE

- Hybrid and electric vehicles may appear to be shut down even when they are not due to the potential lack of engine noise.
- Refer to Chapter 2 “Immobilization” for “non-emergency” and “emergency” to identify the specific shut-down procedure to be followed.

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4. Access to the occupants

4.1 Vehicle shell and body structure

The materials used for the vehicle shell and body structure are shown in Fig. 4.1, including details of high strength steel zones.

Prohibited cut zones are shown in Fig. 4.2.

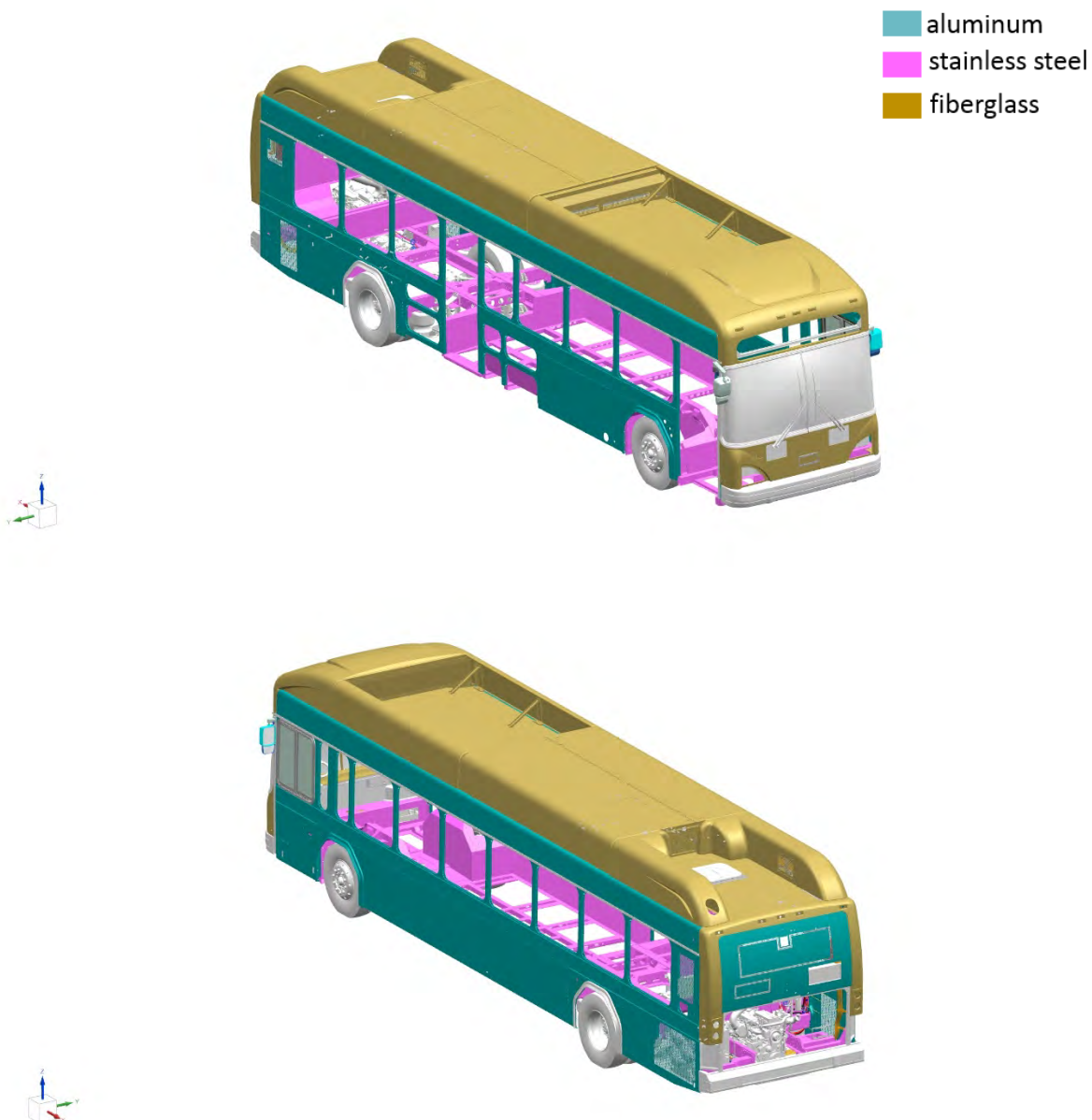


Fig. 4.1. Construction Materials

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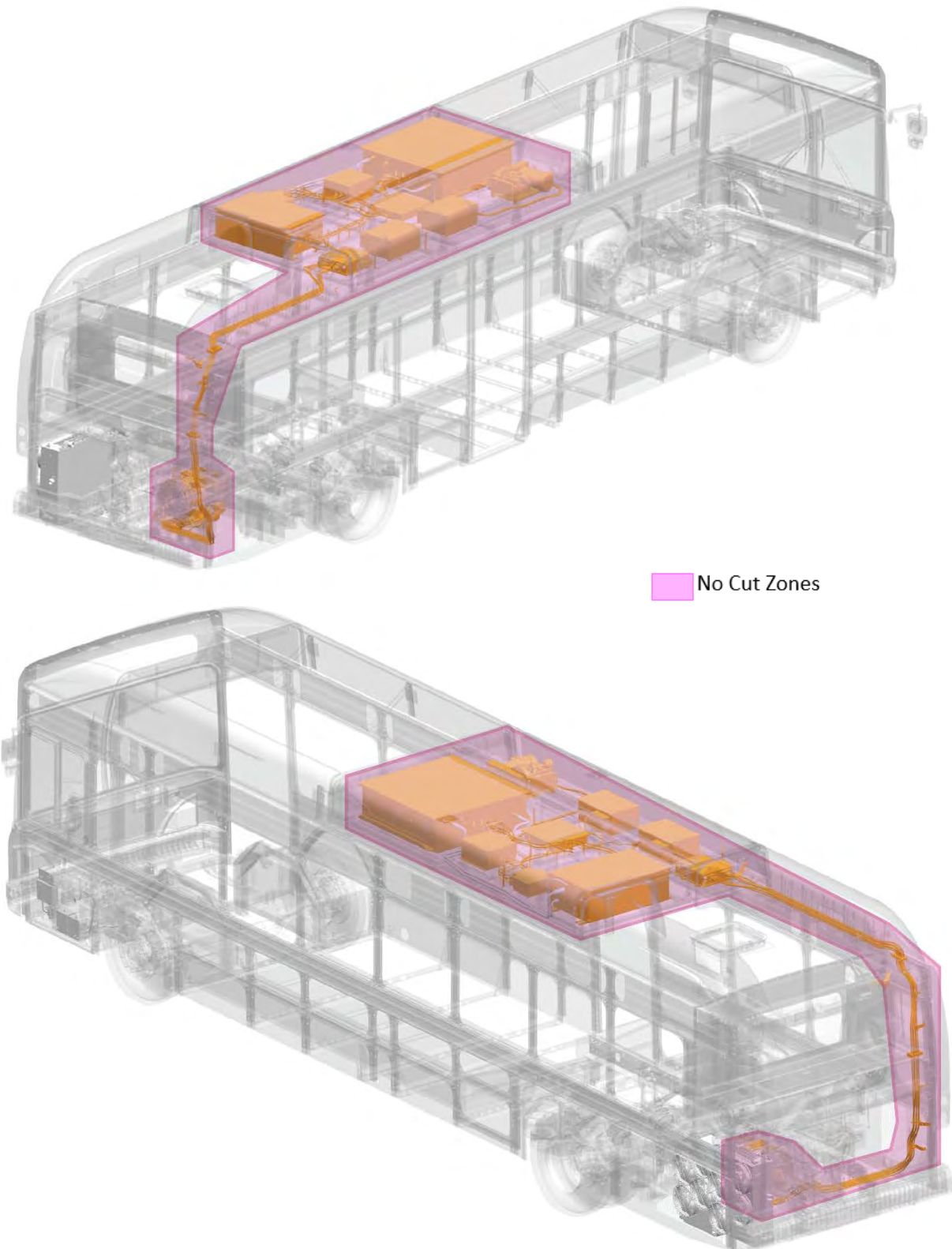
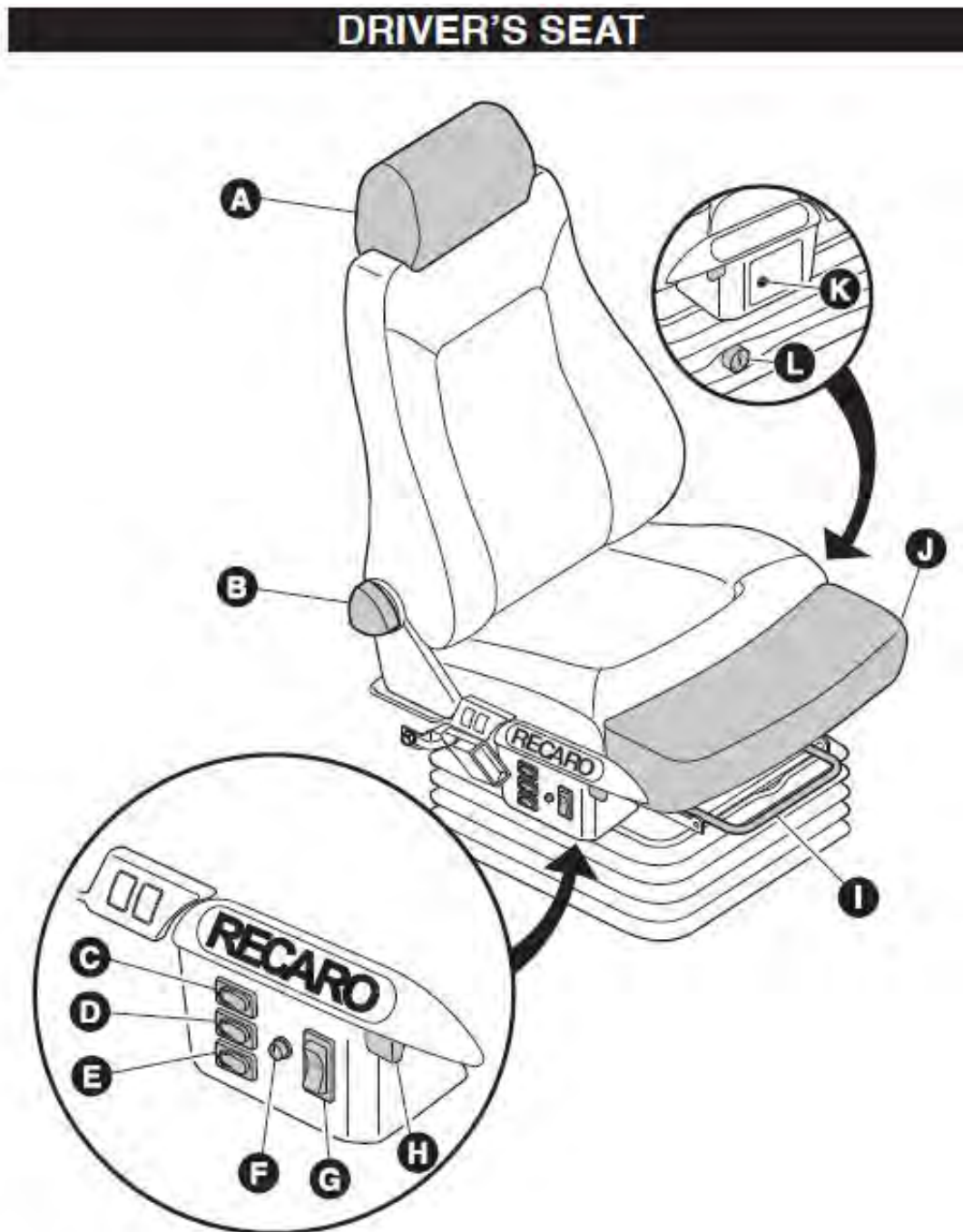


Fig. 4.2 No Cut Zones

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4.2 Adjustment(s) mechanism(s) driver's seat and steering wheel



WARNING

The seat should be adjusted **ONLY** when the bus is stopped, with the transmission in Neutral, and the parking brake applied.

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- A** Adjust the headrest height by pulling up or pushing down to the desired height; adjust the headrest tilt by pulling forward or pushing backward.
- B** The recline angle between the seat back and the cushion can be adjusted using the large knobs located on either side of the seat.
- C** The top switch regulates the pressure in the upper lumbar support cushion. Push the switch forward to inflate the cushion; rearward deflates the cushion.
- D** The middle switch regulates the pressure in the middle lumbar support cushion. Push the switch forward to inflate the cushion; rearward deflates the cushion.
- E** The lower switch regulates the pressure in the lower lumbar support cushion. Push the switch forward to inflate the cushion; rearward deflates the cushion.
- F** Forward and backward adjustment is performed by pushing this button and then sliding the seat to the desired location. Release the button to lock the seat into position.
- G** To raise the level of the seat, press on the upper part of this switch. To lower the seat, depress the lower part of this switch.
- H** The angle of the front seat cushion can be adjusted using the large handles located on either side of the seat base. To adjust the seat cushion angle, pull up on the handles and move the thigh extension cushion up or down to the desired position.
- I** Forward and backward adjustment is performed by pulling up on the handle located at the front base of the seat and then sliding the seat to the desired location. Button **F** performs the same function.
- J** The front part of the seat cushion can be extended for thigh support. Pull the cushion forward or push it back for optimum comfort.
- K** Use this switch to control the optional seat heater.
- L** This seat features an adjustable shock absorber. Turn the knob clockwise to soften the ride or counterclockwise to stiffen the ride.

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STEERING WHEEL

The GILLIG Low Floor features a fully adjustable steering wheel which, in combination with the adjustable driver's seat, permits drivers of all shapes and sizes to comfortably operate the bus.

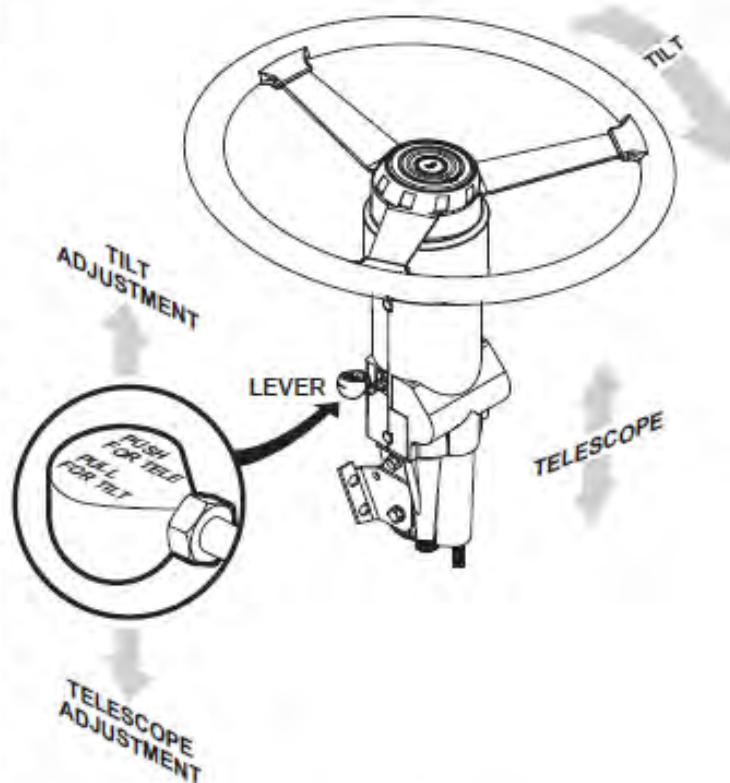


WARNING

The steering wheel should be adjusted only when the bus is stopped, with the transmission in Neutral, and the parking brake applied. Never adjust the steering wheel while the bus is moving!

Both the angle (or tilt) of the steering wheel and the length of the steering column (telescope) can be adjusted. The Steering Column Adjustment Lever, located on the left side of the steering column, controls both of these adjustments.

To adjust the tilt of the steering wheel, pull up on the lever and, while holding the lever up, move the steering wheel to the most comfortable angle. Release the lever to lock the wheel tilt into place.



Steering Column Adjustment

To adjust the length of the telescoping steering column, push down on the lever and, while holding the lever down, lift or push the steering wheel to the proper height. Release the lever to lock the steering column length.

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4.3 Vehicle kneeling system

The kneeling system lowers the front of the bus about 3" below normal ride height to make passenger loading easier. The vehicle has one of the controllers shown in Figure 4.3.



To kneel:

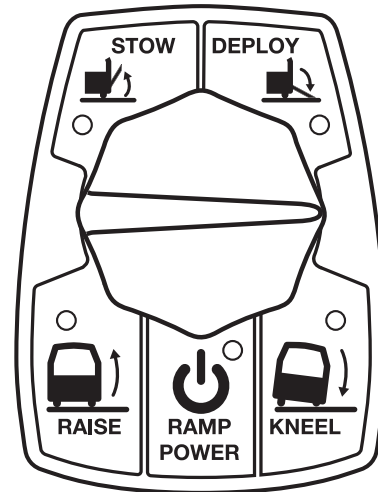
Raise the toggle guard on the switch and hold the toggle in the “KNEEL” position until the bus drops to the correct height.

When the bus kneels, the interlock system disables the throttle and applies the brake interlock.

To raise:

Push the toggle switch to the “RAISE” position and release it immediately.

After 2.5 seconds of raising motion, you can apply the service brake to disengage the interlock system.



To kneel:

Push the Kneel button on the right. The LED on this switch will light and the bus will drop to the correct height.

When the bus kneels, the interlock system disables the throttle and applies the brake interlock.

To raise:

Push the “Raise” button and release it immediately. The LED on this switch will light and the bus will rise to the normal ride height.

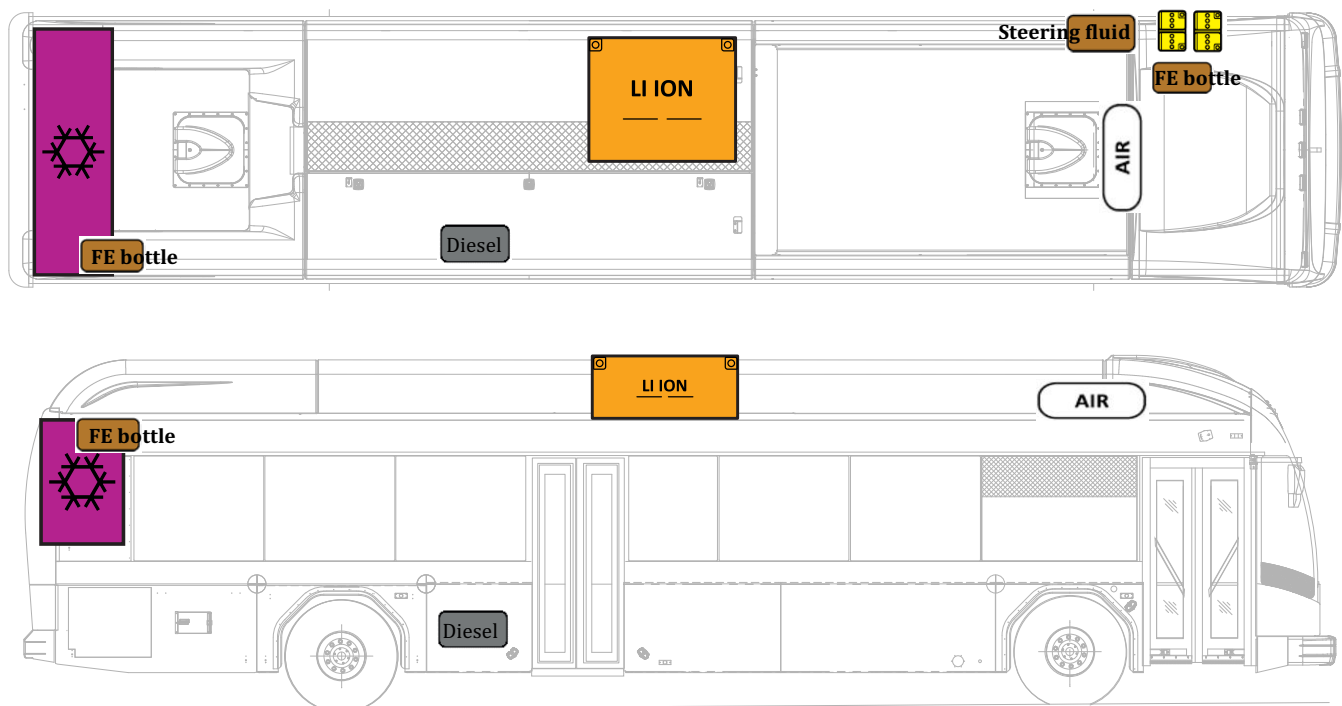
After 2.5 seconds of raising motion, you can apply the service brake to disengage the interlock system.

Figure 4.3. Kneeling System Controllers and Operation

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5. Stored energy / liquids / gases / solids

Component Type	Type/chemistry	Number	Volume/weight	Max. oper. pressure	Specific Danger (Symbol)
RESS Propulsion Batteries	Lithium-Ion	1	1168 lbs (each)		
12/24V Batteries (House)	Lead-acid/AGM	2	75 lbs (each)		
12/24V batteries (Fire suppr. Systems)	Lead-acid/AGM	2	22 lbs		
Air system	Compressed air	4 tanks	6350 cu.in.	130 psi	
Fire suppression bottles	Type ABC powder	2	13 lbs (front/30 lbs (back)	300 psi	
Coolant (ECP, RESS Batteries, HVAC)	Ethylene-glycol/H2O		20 gal	15 psi	
A/C fluid	R407C		5900 g	250 psi	
Flammable fluids/materials					
a) Diesel fuel (FF heater)	Diesel	1 tank	120 gal		
b) Steering fluid	Castrol TranSynd	1 tank	1.1 gal		
c) Rear axle/front wheel bearing fluid	API GL-5/SAE J2360	2 tanks	5.5 gal		
d) Interior materials	In accordance with DOT/FMVSS 302				



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5.1 High-Voltage (HV) battery information — General First Aid measures & environmental aspects

Under normal conditions of use, the battery does not present any risk of exposure to its content. ***However, unforeseen circumstances (e.g., a vehicle collision) may cause damage to one or more battery cells with uncontrolled increases in temperature and pressure (thermal runaway), which can lead to several possible hazards, as described below:***

Exposure to high voltage (>60V):

- Avoid contact with HV cabling and components. ALWAYS assume the HV system is energized.
- Avoid contact with a damaged HV battery, a significant shock hazard may exist.
- NEVER cut orange HV cabling or penetrate HV components with tools.
- HV system shutdown procedures in Chapter 2 are designed to disable the vehicle's HV system, not to discharge the HV battery. THE HV BATTERY WILL REMAIN ENERGIZED.
- Even with the HV battery completely discharged (State of charge = 0), the system remains within the class B voltage definition and should therefore be disconnected as indicated in Chapter 2 "Immobilization".
- Sparks, smoke, or bubbling noises coming from the HV battery are signs of a potentially overheating battery, which could result in a delayed fire.
- Follow local medical protocols and First Aid SOPs for any burn, electrical, or other injuries.

Exposure to electrolyte mixture:

- Wear appropriate PPE if exposure to electrolyte is expected. SCBA is highly recommended due to the possibility of severely irritating fumes.
- Any clothing or PPE that may have come into contact with electrolyte should be either decontaminated or discarded appropriately.

Inhalation in non-fire situations:

- If you detect leaking fluids, sparks, smoke, or bubbling noises coming from the HV battery, ventilate the vehicle by opening the windows to prevent the buildup of fumes.
- If electrolyte leaks and gets exposed to the air, electrolytic vapors may be released. Even in a non-fire situation, the electrolytic vapors may be toxic or at least severely irritating. If vapors are inhaled, immediately move to fresh air.

Safety measures for battery handling:

- The battery assembly cover should never be breached or removed under any circumstances, including fire. Doing so might result in severe electrical burns, shocks, or electrocution.

Harmful and/or flammable fumes:

- Contents of HV battery should be considered corrosive, toxic and/or flammable
- If you detect unusual odors or experience eye, nose, throat, or skin irritation, use full PPE with SCBA.
- If you detect leaking fluids, sparks, smoke, or bubbling noises coming from the HV battery, ventilate the vehicle by opening doors and windows to prevent the buildup of fumes.
- Sparks, smoke, or bubbling noises coming from the HV battery are signs of a potentially overheating battery, which could result in a delayed fire.

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Spill/leak hazards from HV batteries:

- HV Li-Ion batteries are considered dry cell batteries, and if damaged or breached, electrolyte leakage should be minimal.
- The HV battery of this vehicle is liquid cooled. If the battery is damaged, coolant may leak. The coolant is a water/glycol/ethylene solution, similar to that of the conventional vehicle radiators and should not be confused with battery electrolyte.
- If damage is extensive, cross contamination with battery electrolyte is possible.

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6. In case of fire

6.1 General information

This vehicle is equipped with a fire detection system and automatic shutdown. When excessive temperatures are detected in the powertrain or HVAC compartments, an automatic shutdown is initiated and will occur unless overridden by the operator.

An automatic shutdown will:

- Turn off high voltage and the diesel engine, removing propulsion power and bringing the vehicle to a stop
- Keep low voltage turned on, and the power steering pump will continue to be energized and running

Low voltage can be turned off as indicated in Section 2B Emergency Immobilization Procedure.

In case of fire, use standard vehicle firefighting equipment and tactics in accordance with department standard operating procedures SOPs/SOGs.

Hybrid-electric vehicles do not require special equipment for fire suppression and/or extinguishment.

6.2 Battery fire situations

- Establish a 20-foot radius “safety zone” around the vehicle.
- Use abundant amounts of water to extinguish a battery fire. Do not use ABC fire extinguishers to put out a battery fire.
- DO NOT EXTINGUISH BATTERY FIRES WITH SMALL AMOUNTS OF WATER. Establish an additional water source as battery fires can take up to 24-hours to burn out and several thousand gallons of water.
- Use a thermal imaging camera to ensure all heat sources are extinguished and prevent from risk of reignition.
- Standard firefighting practices can be applied to other portions of the vehicle to prevent the fire from spreading.

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7. In case of submersion

- The high-voltage system is isolated from the chassis and is designed to pose no shock hazard from touching the vehicle body.
- The system is designed not to energize surrounding water and is equipped with short-circuit fault detectors designed to shut down the high-voltage system in the event of a short.
- On a submerged vehicle, avoid contact with HV components, cabling, or service disconnects.
- Follow standard departmental practices and procedures for patient access and vehicle removal from water.

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8. Towing / transportation / storage

8.1 Vehicle removal information

- GILLIG recommends flat bedding a disabled bus.
- If towing is needed, flat towing from the front with a fixed tow bar is recommended.
- Towing with the front end raised should be a last resort. GILLIG does not recommend towing from the rear.
- For complete flat bedding and towing procedures, refer to the operator's manual included in this vehicle
- **WARNING: Remove the driveshaft or *both* axle shafts before towing the bus! Towing with the driveline rotating may create risks of FIRE in the High-Voltage electrical system.**

8.2 Battery information

- In a post fire/crash situation or if damage to the battery is observed or suspected, immediately notify the manufacturer indicated on the "Identification" tags in Chapter 1;
- Notify tow operators removing the vehicle of the need of trained personnel to inspect it and provide recommendations for storage and disposal.
- Due to the potential of delayed fire, do not store a severely damaged vehicle containing an HV Lithium-Ion battery in or within 150 feet of a structure or other vehicle.
- Should removal of the battery pack be needed, refer to the procedure specified in the vehicle operator's manual included in this vehicle.

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9. Important additional information

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10. Explanation of pictograms used

The text and images on the following pages have been copied from **Annex B of ISO 17840-3:2019(E)** *Road vehicles – Information for first and second responders – Parts 3: Emergency response guide template*.

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Annex B
(normative)

Pictograms for use in ISO 17840

Components/functions/actions that shall be considered during the rescue procedure are represented by dedicated pictograms. The pictograms are used:

- to indicate the location of the respective components/functions in the vehicle, in conjunction with the rescue sheet illustration (for details, see ISO 17840-1 and ISO 17840-2);
- to communicate a specific function or danger, for use under the rescue sheet additional pages headings and ERG headings;
- to communicate the recognition of propulsion type; and
- to indicate the extinguish measures.

Level of importance:

- 1 = Crucial information for the rescue operations, as applicable to the vehicle type/model; and
- 2 = Optional information, to further assist the rescue procedures.

Tables B.1 to B.8 list the pictograms for the components and functions to be considered.

NOTE Where applicable, the pictograms are shown both with and without the corner marks for the basic symbol pattern according to IEC 80416-1.

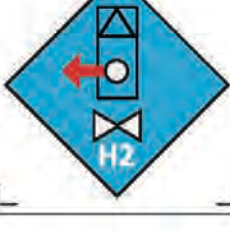
Table B.1 — Pictograms concerning recognition

Pictogram	Designation and remarks
	Examples of propulsion type recognition Reference: ISO 17840-4 Level of importance: 1 To be used in/on: — Rescue sheet illustration; — ERG under heading 1. NOTE Pictogram examples for fuel cell and hybrid electric Diesel propulsion are shown. See ISO 17840-4 for principles and other propulsion pictograms.

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Table B.1 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Direction of overpressure valve Function/description: To indicate the direction of the overpressure valve. Direction arrow in red colour = direction overpressure valve "forward, back, left, right". Red dot in black circle = direction overpressure safety valve "up". Red hatched circle = direction overpressure safety valve "down". Level of importance: 1 Reference: — ISO 17840-1 — Application of ISO 7000-0234 Colours: — Red, RGB: 255,0,0 — Black — Background pictogram: — Hydrogen: Light blue, RGB: 0,176,240 — Gas: Green, RGB: 0,176,80 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5. NOTE UN-ECE R.110 only specifies upward discharge direction (for LNG and CNG).
	
	
	
	

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Table B.1 (continued)

Pictogram	Designation and remarks
	
	
	
	
	

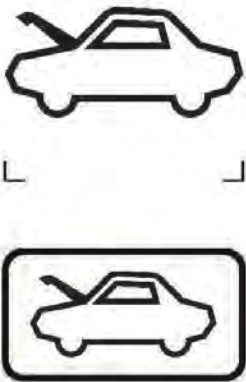
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Table B.1 (continued)

Pictogram	Designation and remarks
	

Table B.2 — Pictograms concerning access to the components

Pictogram	Designation and remarks
	Title/Meaning/Referent: Bonnet; hood Function/description: To identify the control that opens the compartment located outside the passenger area in the front of the vehicle. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-0241 To be used in/on: — Rescue sheet illustration; — ERG under heading 3.

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Table B.2 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Boot; Trunk Function/description: To identify the control that opens the compartment located outside the passenger area in the rear of the vehicle. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-0242 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
	Title/Meaning/Referent: Hood release, truck Function/description: To identify the control for the hood release in a conventional heavy truck. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-2962 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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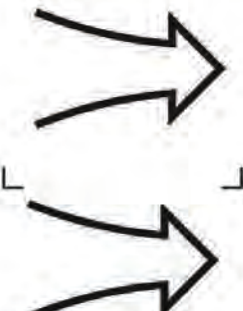
Table B.3 — Pictograms concerning disabling of the vehicle (excluding high voltage)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Device to shut down power in vehicle Function/description: Shutdown power in vehicle, in all forms, by means of: — Ignition key; — Push button; — Operation in engine compartment; — Operation on dashboard; — Battery switch; — Other. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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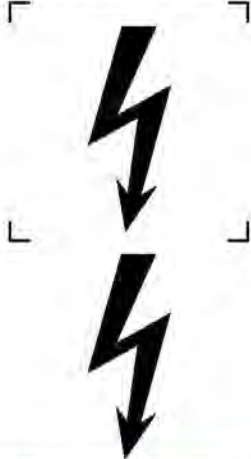
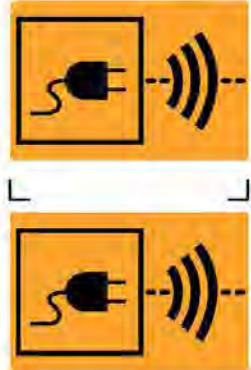
Table B.3 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Remove smart key /starter key. Function/description: To indicate that the smart key should be removed from the vehicle to prevent accidental starting of the vehicle. A safe distance may optionally be indicated. Level of importance: 1 Colours: — Black — White Reference: — Application of ISO 7000-2849; — Application of ISO 7000-2617. To be used in/on: — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
	Title/Meaning/Referent: Air intake Function/description: To identify the air intake where CO₂ can be blown to stop the engine. Level of importance: 1 Colours: Black Reference: ISO 7000-1604 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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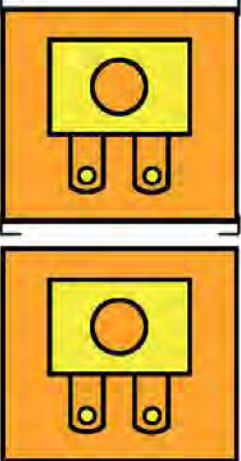
Table B.4 — Pictograms concerning disabling of the vehicle high voltage (EV, HEV, PHEV, FCEV)

Pictogram	Designation and remarks
— Orange = High Voltage (Class B Voltage) — Yellow = Controlling the High Voltage by Low Voltage — Orange frame = Procedure for disabling High Voltage vehicle	
	Title/Meaning/Referent: Dangerous voltage Function/description: To indicate hazards arising from dangerous voltages. Level of importance: 1 Colours: Black Reference: IEC 60417-5036 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading where needed; — ERG under heading where needed.
	Title/Meaning/Referent: Vehicle induction charging Function/description: To indicate the vehicle is connected to an electromagnetic induction source for charging the high voltage traction batteries. To identify the location of the induction charging system or components. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: Application of ISO 7000-2616 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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Table B.4 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Fuse box disabling high voltage Function/description: To identify the low voltage fuse that controls the high voltage. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration and secondary pages under heading 3. — ERG under heading 3. NOTE This is a method used by various car manufacturers.
	Title/Meaning/Referent: Cable cut Function/description: To identify the cable to cut that disconnect high voltage and SRS components. To show that two separate places in the same cable shall be cut. Size and proportions can be adjusted to fit the intended purpose. Level of importance: 1 Colours: — Yellow, RGB: 255,255, — Orange, RGB: 255,165,0 — Red, RGB: 206,17,38 — Black Reference: — To be used in/on: — Rescue sheet illustration and additional pages under heading 3; — ERG under heading 3. NOTE This is a method used by various car manufacturers.

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Table B.4 (continued)

Pictogram	Designation and remarks
 	Title/Meaning/Referent: Disconnect high voltage device (e.g. service plug) Function/description: To identify the high voltage device that disconnects the high voltage, where appropriate PPE is needed for the action. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
 	Title/Meaning/Referent: Disconnect high voltage device Function/description: To identify the low voltage device that disconnects the high voltage. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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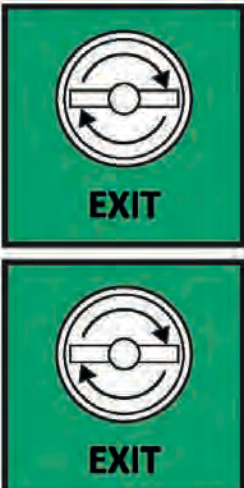
Table B.5 — Pictograms concerning access to the occupants

Pictogram	Designation and remarks
 	Title/Meaning/Referent: Emergency exit right hand (emergency hatches, ...) Function/description: To indicate an escape route to a place of safety. Level of importance: 1 Colours: — Green, RGB: 0,160,104 — White Reference: ISO 7010-E002 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 4; — ERG under heading 4.
 	Title/Meaning/Referent: Emergency exit left hand (emergency hatches, ...) Function/description: To indicate an escape route to a place of safety. Level of importance: 1 Colours: — Green, RGB: 0,160,104 — White Reference: ISO 7010-E001 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 4; — ERG under heading 4.

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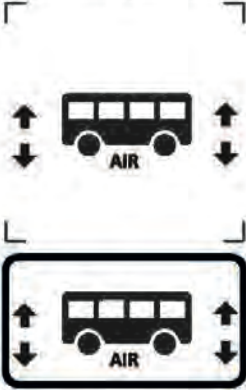
Table B.5 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Break to obtain access Function/description: To indicate a cover which requires breaking to obtain access to an emergency exit device. Level of importance: 1 Colours: — Green, RGB: 0,160,104 — White Reference: ISO 7010-E008 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 4; — ERG under heading 4.
	Title/Meaning/Referent: Emergency door opener Function/description: To indicate the emergency switch to open doors. Level of importance: 1 Colours: — Green, RGB: 0,160,104 — White — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 4; — ERG under heading 4.

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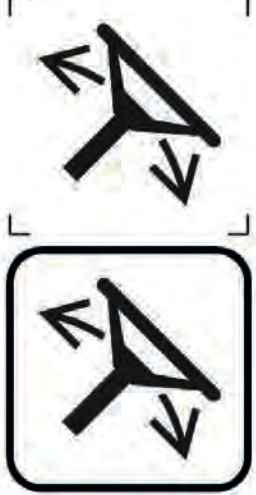
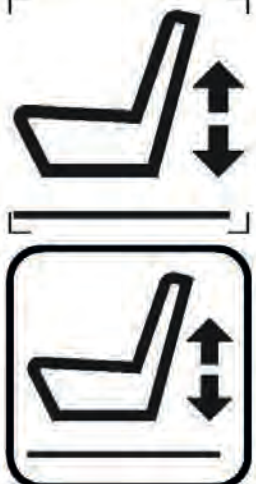
Table B.5 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Height control bus, by air system Function/description: To identify the control that moves the chassis upward or downward by air. A frame may be used to separate the pictogram from the background as needed. Level of importance: 1 Colours: — White — Black Reference: — Application of ISO 7001-PI TF 006 — Application of ISO 7000-2462 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 2; — ERG under heading 2.
	Title/Meaning/Referent: Height control truck, by air system Function/description: To identify the control that raises or lowers the entire vehicle by air. A frame may be used to separate the pictogram from the background as needed. Level of importance: 1 Colours: — Black — White Reference: Application of ISO 7000-2461 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 2; — ERG under heading 2.

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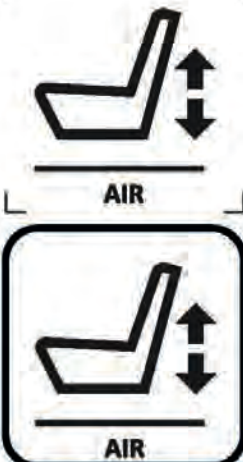
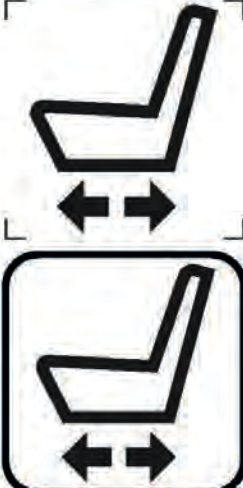
Table B.5 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Steering wheel, tilt control Function/description: To identify the control that allows adjustment of the steering wheel by tilting up or down. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-2064 To be used in/on: — ERG under heading 4
	Title/Meaning/Referent: Seat height adjustment Function/description: To identify the control that moves the entire seat upward or downward. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-1430 To be used in/on: — ERG under heading 4.

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Table B.5 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Seat height adjustment by air system Function/description: To identify the control that moves the entire seat upward or downward by air. A frame may be used to separate the pictogram from the background as needed. Level of importance: 1 Colours: — Black — White Reference: Application of ISO 7000-1430 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 4; — ERG under heading 4.
	Title/Meaning/Referent: Seat adjustment, longitudinal Function/description: To identify the control that moves the entire seat forward or rearward. A frame may be used to separate the pictogram from the background as needed. Level of importance: 2 Colours: — Black — White Reference: Application of ISO 7000-1428 To be used in/on: — ERG under heading 4.

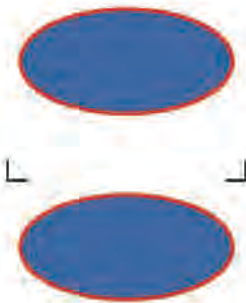
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Table B.5 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Lifting point; central support Function/description: To identify the locations on the equipment where a lifting jack or support device can be used. Level of importance: 1 Colours: — Black — White Reference: ISO 7000-0542 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 2; — ERG under heading 2.

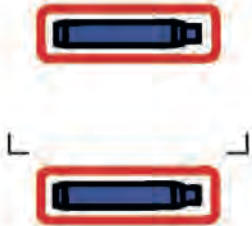
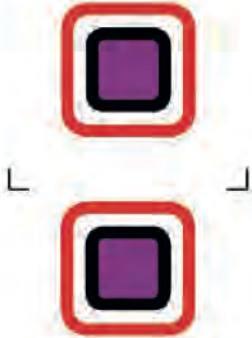
Table B.6 — Other vehicle related pictograms

Pictogram	Designation and remarks
	Title/Meaning/Referent: Airbag Function/description: To identify an airbag. Level of importance: 1 Colours: — Blue, RGB: 77,77,255 — Red, RGB: 255,0,0 Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Remarks: Pictogram can be adjusted to represent the actual size and form. Different types of airbag-related occupant protection systems can be shown using the airbag pictogram with an appropriate size and form, e.g.: — side airbag; — curtain airbag; — knee airbag; — inflatable seat belt.

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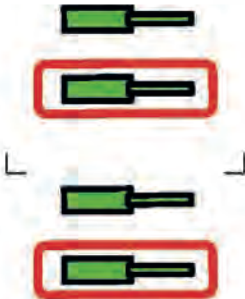
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Airbag inflator/stored gas inflator Function/description: To identify an airbag inflator/stored gas inflator. Level of importance: 1 Colours: — Blue, RGB: 77,77,255 — Red, RGB: 255,0,0 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Remarks: Pictogram can be adjusted to represent the actual size and form. Pictogram is used to show the location of the stored gas inflator for e.g. inflatable curtains or pedestrian protection active system. This pictogram should not be shown for conventional airbag systems with integrated gas inflator, such as frontal airbag in the steering wheel or in the dashboard, side airbag, knee airbag.
	Title/Meaning/Referent: Seat belt pretensioner Function/description: To identify a seat belt pretensioner. Level of importance: 1 Colours: — Purple, RGB: 152,43,143 — Red, RGB: 255,0,0 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Remarks: If a seating position has more than one pretensioner (e.g. for lap and shoulder belt), each pretensioner location shall be indicated by pictogram. Pictogram can be adjusted to represent the actual size and form. It can also be a combination of simple forms.

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ISO 17840-3:2019(E)

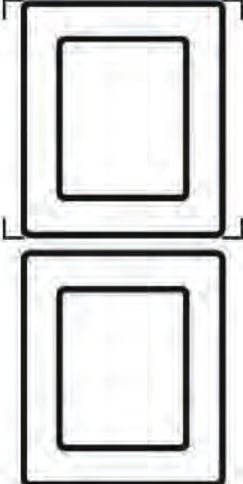
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Gas strut, preloaded spring Function/description: To identify a gas strut, preloaded spring. Level of importance: 1 Colours: Remarks: — Lime green, RGB: 0,255,0 — Red, RGB: 255,0,0 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Red surrounding is used only if the device is triggered. Pictogram can be adjusted to represent the actual size and form.
	Title/Meaning/Referent: Pedestrian protection active system Function/description: To identify the pedestrian protection active system. Level of importance: 1 Colours: — Red RGB: 255/0/0; — Black; — White. Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Remarks: Pictogram for pedestrian protection active system shall be used to inform that the vehicle is equipped with a system that can deploy, e.g. the bonnet/hood. The pictogram background is white by default but can alternatively be using the colour of the activation mechanism. The pictogram can be combined with or connected to the activation mechanism (airbag, gas inflator, gas strut, preloaded spring) for deploying the system, e.g. the bonnet/hood.

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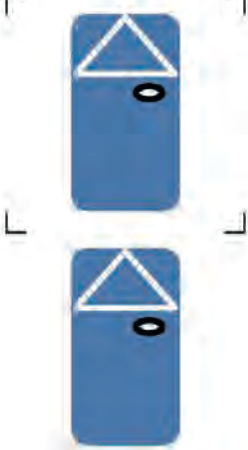
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: High strength zone Function/description: To identify a high strength zone. Level of importance: 1 Colours: — Sea green, RGB: 0,128,128 — Black — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9. Remarks: Pictogram can be adjusted to represent the actual size and form.
	Title/Meaning/Referent: Zone requiring special attention Function/description: To identify the zone requiring special attention. Level of importance: 1 Colours: — Black — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration and secondary pages under heading 5; — ERG under heading 5. Remarks: Pictogram can be adjusted to represent the actual size and form.

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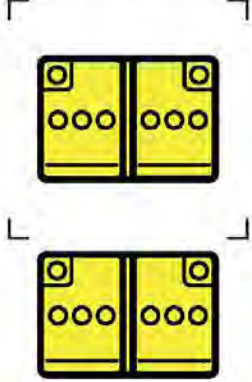
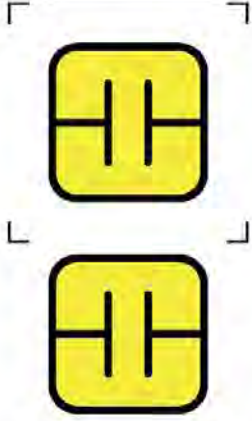
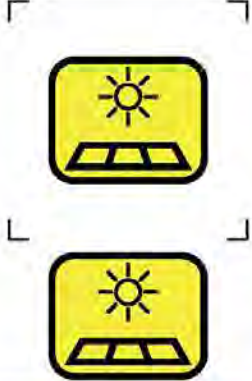
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Carbon structure Function/description: To indicate that carbon is used in the chassis structure. To inform about risks of inhalation, appropriate PPE is needed. Level of importance: 1 Colours: — Black — White Reference: — To be used in/on: — Rescue sheet secondary pages under heading 5; — ERG under heading 5.
	Title/Meaning/Referent: Left hand drive Function/description: To identify a left-hand drive vehicle. Level of importance: 1 Colours: — Blue, RGB: 79,129,189 — Black — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration. Remarks: For use in the header of the rescue sheet. The colour can be adjusted to contrast with the background of the header.
	Title/Meaning/Referent: Right hand drive Function/description: To identify a right-hand drive vehicle. Level of importance: 1 Colours: — Blue, RGB: 79,129,189 — Black — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration. Remarks: For use in the header of the rescue sheet. The colour can be adjusted to contrast with the background of the header.

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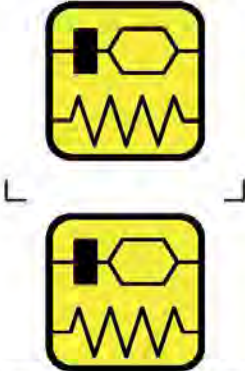
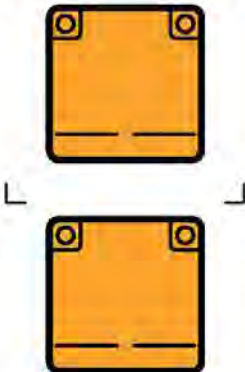
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Battery, low-voltage Function/description: To identify a low voltage battery. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5. Remarks: For class A voltage application. It shall be accompanied with the technology of the battery (e.g. Li-Ion or Ni-MH) if different from a conventional battery type.
	Title/Meaning/Referent: Ultra-capacitor, low-voltage Function/description: To identify a low voltage ultra-capacitor. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5. Remarks: For class A voltage application.
	Title/Meaning/Referent: Solar Panel Function/description: To identify a solar panel. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages; — ERG under heading 3.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: SRS control unit Function/description: To identify a SRS control unit. Level of importance: 1 Colours: — Yellow, RGB: 255,255,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 9.
	Title/Meaning/Referent: Battery pack, high-voltage Function/description: To indicate a high voltage battery pack Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3. Remarks: Pictogram can be adjusted to represent the actual size and form. It shall be accompanied with the technology of the battery (e.g. Li-Ion or Ni-MH). Optionally, the nominal voltage value of the battery may be added.

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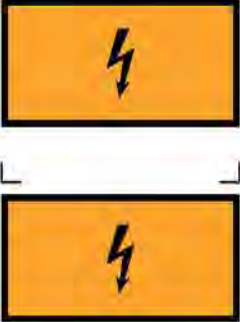
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Fuel cell component Function/description: To indicate a fuel cell component. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Light blue, RGB: 0,176,240 — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5. Remarks: Pictogram can be adjusted to represent the actual size and form.
	Title/Meaning/Referent: High voltage ultra-capacitor Function/description: To indicate an ultra-capacitor pack. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3. Remarks: For class B voltage application.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: High voltage component Function/description: To indicate a high voltage component. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3. — ERG under heading 3. Remarks: For class B voltage application. Pictogram can be adjusted to represent the actual size and form. Flash may be omitted in case of space constraints.
	Title/Meaning/Referent: High voltage power cable Function/description: To identify a high voltage power cable. Level of importance: 1 Colours: Orange, RGB: 255,165,0 Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 3. Remarks: For class B voltage application. Pictogram can be adjusted to represent the actual shape and cable path. It can optionally have a black contour line. HV components should be possible to differentiate from HV battery pack. Legend and pictogram graphics should correspond with regard to the use of contour line concept.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Fuel tank content Diesel Function/description: To indicate the content of the tank by using a defined colour. Level of importance: 1 Colours: — Grey, RGB: 127,127,127 — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5.
	Title/Meaning/Referent: Fuel tank content gasoline/ethanol Function/description: To indicate the content of the tank by using a defined colour. Level of importance: 1 Colours: — Dark red, RGB: 139,0,0 — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5.
	Title/Meaning/Referent: Tank content oil (e.g. hybrid oil technology) Function/description: To indicate the content of the tank by using a defined colour. Level of importance: 1 Colours: — Brown RGB: 183,120,29 — Black Reference: — To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Gas tank with gas type indication (CNG) Function/description: To indicate the content of the tank by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 5.
	Title/Meaning/Referent: Manual gas shut-off valve with gas type indication (CNG) Function/description: To indicate the manual gas shut-off valve by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Automatic gas overpressure safety valve with gas type indication (CNG) Function/description: To indicate the device that controls gas overpressure in a tank by using a defined colour and indication of gas type. — Controlled by pressure (pressure release device); — Controlled by temperature (temperature pressure release device). Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
	Title/Meaning/Referent: Gas tank with gas type indication (LPG) Function/description: To indicate the content of the tank by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 5.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Manual gas shut-off valve with gas type indication (LPG) Function/description: To indicate the manual gas shut-off valve by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
	Title/Meaning/Referent: Automatic gas overpressure safety valve with gas type indication (LPG) Function/description: To indicate the device that controls gas overpressure in a tank by using a defined colour and indication of gas type. — Controlled by pressure (pressure release device); — Controlled by temperature (temperature pressure release device). Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Gastank with gas type indication (DME) Function/description: To indicate the content of the tank by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green RGB: 0,176,80 — White. Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 5.
	Title/Meaning/Referent: Manual gas shut-off valve with gas type indication (DME) Function/description: To indicate the manual gas shut-off valve by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Automatic gas overpressure safety valve with gas type indication (DME) Function/description: To indicate the device that controls gas overpressure in a tank by using a defined colour and indication of gas type. - Controlled by pressure (pressure release device); - Controlled by temperature (temperature pressure release device). Level of importance: 1 Colours: - Green RGB: 0,176,80 - White Reference: - ISO 17840-1; - Application of ISO 7000-1852. To be used in/on: - Rescue sheet illustration; - Rescue sheet secondary pages under heading 3; - ERG under heading 3.
	Title/Meaning/Referent: Gas tank with gas type indication (LNG) Function/description: To indicate the content of the tank by using a defined colour and indication of gas type. Level of importance: 1 Colours: - Green, RGB: 0,176,80 - White Reference: ISO 17840-1 To be used in/on: - Rescue sheet illustration; - Rescue sheet secondary pages under heading 5; - ERG under heading 5.

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Table B.6 (continued)

Pictogram	Designation and remarks
 	Title/Meaning/Referent: Manual gas shut-off valve with gas type indication (LNG) Function/description: To indicate the manual gas shut-off valve by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
 	Title/Meaning/Referent: Automatic gas overpressure safety valve with gas type indication (LNG). Function/description: To indicate the device that controls gas overpressure in a tank by using a defined colour and indication of gas type. — Controlled by pressure (pressure release device); — Controlled by temperature (temperature pressure release device). Level of importance: 1 Colours: — Green, RGB: 0,176,80 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.

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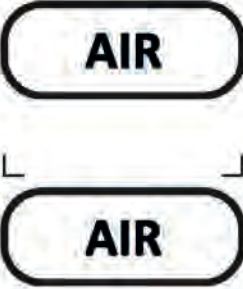
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Gas tank with gas type indication (H₂) Function/description: To indicate the content of the tank by using a defined colour and indication of gas type. Level of importance: 1 Colours: — Light blue RGB: 0,176,240 — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 5.
	Title/Meaning/Referent: Manual gas shut-off valve with gas type indication (H₂). Function/description: To indicate the manual gas shut-off valve by using a defined colour for hydrogen and indication of gas type. Level of importance: 1 Colours: — Light blue, RGB: 0,176,240 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration and secondary pages under heading 3; — ERG under heading 3.

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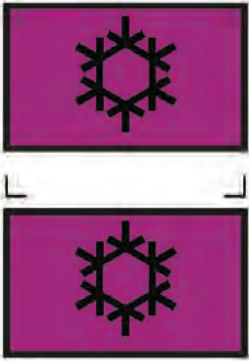
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Automatic hydrogen overpressure safety valve with gas type indication Function/description: To indicate the device that controls gas overpressure in a tank by using a defined colour and indication of gas type. — Controlled by temperature (temperature pressure release device). Level of importance: 1 Colours: — Light blue RGB: 0,176,240 — White Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 3; — ERG under heading 3.
	Title/Meaning/Referent: Air tank Function/description: To indicate an air tank. Level of importance: 1 Colours: — Black — White Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 5.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Air-conditioning component Function/description: To indicate an air conditioning component by using a defined colour. Type of coolant shall be mentioned in additional pages and rescue sheet (e.g. CO₂, fluor-carbon based chemistry). Level of importance: 1 Colours: — Purple, RGB: 204,0,204 — Black Reference: Application of ISO 7000-0027 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 5; — ERG under heading 5 Remarks: Pictogram can be adjusted to represent the actual size and form. Snow flake may be omitted in case of space constraints.
	Title/Meaning/Referent: Gas line (generic) Function/description: To indicate a gas line by using a defined colour. Level of importance: 1 Colours: Green, RGB: 0,176,80 Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 5. Remarks: Pictogram can be adjusted to represent the actual shape and line path.

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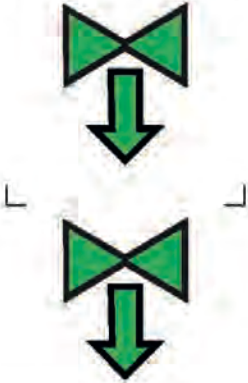
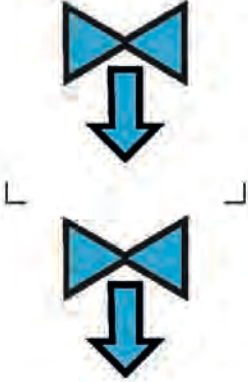
Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Gas line (H₂) Function/description: To indicate a gas line for hydrogen by using a defined colour. Level of importance: 1 Colours: Light blue, RGB: 0,176,240 Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 5. Remarks: Pictogram can be adjusted to represent the actual shape and line path.
	Title/Meaning/Referent: Air-conditioning line Function/description: To indicate an air-conditioning line by using a defined colour. Type of coolant or name shall be mentioned (e.g. CO₂, fluor-carbon based chemistry). Level of importance: 1 Colours: Purple, RGB: 204,0,204 Reference: ISO 17840-1 To be used in/on: — Rescue sheet illustration; — ERG under heading 5. Remarks: Pictogram can be adjusted to represent the actual shape and line path.

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Table B.6 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Direction gas overpressure safety valve (e.g. LPG) in vehicle Function/description: To indicate the direction gas overpressure valve on an illustration by using a defined colour. Level of importance: 1 Colours: — Green, RGB: 0,176,80 — Black Reference: — ISO 17840-1; — Application of ISO 7000-0234; — Application of ISO 7000-0251. To be used in/on: — Rescue sheet illustration.
	Title/Meaning/Referent: Direction hydrogen overpressure safety valve in vehicle Function/description: To indicate the direction of the hydrogen overpressure safety valve on an illustration by using a defined colour. Level of importance: 1 Colours: — Light blue, RGB: 0,176,240 — Black Reference: — ISO 17840-1; — Application of ISO 7000-1852. To be used in/on: — Rescue sheet illustration

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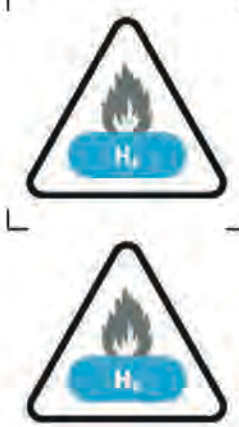
Table B.7 — Pictograms related to fire fighting and safety

Pictogram	Designation and remarks
 	Title/Meaning/Referent: General warning sign Function/description: To signify a general warning. Level of importance: 1 Colours: — Yellow, RGB: 243,200,54 — Black Reference: ISO 7010-W001 To be used in/on: — Rescue sheet secondary pages under heading where necessary; — ERG under heading where necessary.
 	Title/Meaning/Referent: Warning, Electricity Function/description: To warn of electricity and dangerous voltage. Level of importance: 1 Colours: — Yellow, RGB: 243,200,54 — Black Reference: ISO 7010-W012 To be used in/on: — Rescue sheet secondary pages under heading where necessary; — ERG under heading where necessary.
 	Title/Meaning/Referent: Warning, low temperature Function/description: To indicate the dangers concerning low temperatures, e.g. frost bites due to cold gas (e.g. LNG, air-conditioning gas) Level of importance: 1 Colours: — Yellow, RGB: 243,200,54 — Black Reference: ISO 7010-W010 To be used in/on: — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.

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Table B.7 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Attention; hydrogen burns with an almost invisible flame Function/description: To indicate the danger of almost invisible hydrogen flames. Level of importance: 1 Colours: — Light blue, RGB: 0,176,240 — Black — White — Grey Reference: — To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.
	Title/Meaning/Referent: Use thermal Infrared camera Function/description: To indicate that a thermal infrared camera should be used to detect a fire. Level of importance: 2 Colours: — Black — White Reference: — IEC 60878 — Application of IEC 60417:5116 — Application of IEC 60417:6151 — Application of ISO 14617-13-2807 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.

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Table B.7 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Automatic fire suppression system Function/description: To indicate an automatic fire suppression system in the vehicle. Level of importance: 1 Colours: — Red, RGB: 202,17,18 — Black — White Reference: Application of ISO 7010-F001 To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under heading 6; — ERG under heading 6.
	Title/Meaning/Referent: Special battery access Function/description: Special access to put water in the HV battery of an EV vehicle. Level of importance: 1 Colours: — Orange, RGB: 255,165,0 — Black — White Reference: — Application of ISO 7000-2418 — Application of IEC 60417-6171 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.

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Table B.7 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Use water to extinguish the fire Function/description: To indicate that water shall be used to extinguish the fire. Level of importance: 1 Colours: — Blue, RGB: 0,83,135 — White Reference: Application of ISO 7010-P011 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.
	Title/Meaning/Referent: Use wet foam to extinguish the fire Function/description: To indicate that wet foam shall be used to extinguish the fire. System in which a foam concentrate and air are continuously added under pressure to the water being discharged from a fire-fighting pump (CAFS). Wet foam operation is defined by a nominal foam solution/air volume ratio between 1:3 and 1:10, being mixed in the CAFS. Level of importance: 1 Colours: — Blue, RGB: 0,83,135 — White Reference: Application of ISO 7000-3309 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.

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Table B.7 (continued)

Pictogram	Designation and remarks
 	Title/Meaning/Referent: Use dry foam to extinguish the fire Function/description: To indicate that dry foam shall be used to extinguish the fire. System in which a foam concentrate and air are continuously added under pressure to the water being discharged from a fire-fighting pump (CAFS). Dry foam operation is defined by a nominal foam solution/air volume ratio greater than 1:10, being mixed in the CAFS. Level of importance: 1 Colours: — Blue, RGB: 0,83,135 — White Reference: Application of ISO 7000-3309 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.
 	Title/Meaning/Referent: Use ABC powder to extinguish the fire Function/description: To indicate that ABC powder shall be used to extinguish the fire. Level of importance: 1 Colours: — Blue RGB: 0,83,135 — White Reference: — ISO 7202; — Application of ISO 7000-2820; — Application of ISO 7000-3309. To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.

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Table B.7 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Do not extinguish with water Function/description: To prohibit using water to extinguish a fire. Level of importance: 1 Colours: — Red, RGB: 183,31,46 — Black — White Reference: ISO 7010-P011 To be used in/on: — Rescue sheet secondary pages under heading 6; — ERG under heading 6.

Table B.8 — Globally harmonized symbols

Pictogram	Designation and remarks
	Title/Meaning/Referent: Explosive Function/description: To indicate the risk of an explosion. Level of importance: 1 Colours: — Red, RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet secondary pages under headings 5, 6, 8, 9; — ERG under headings 5, 6, 8, 9.

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Table B.8 (continued)

Pictogram	Designation and remarks
 	Title/Meaning/Referent: Flammable Function/description: To indicate the risk of flammability. Level of importance: 1 Colours: — Red RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet secondary pages under headings 5, 6, 8, 9; — ERG under headings 5, 6, 8, 9.
 	Title/Meaning/Referent: Gases under pressure Function/description: To indicate the risk of gases under pressure. Level of importance: 1 Colours: — Red, RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.

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Table B.8 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Oxidizer Function/description: To indicate the risk of oxidizing material/substances. Level of importance: 1 Colours: — Red RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.
	Title/Meaning/Referent: Corrosives Function/description: To indicate the risk of corrosive material/substances. Level of importance: 1 Colours: — Red RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.

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Table B.8 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Hazardous to the human health Function/description: To indicate the risk of damaging human health. Level of importance: 1 Colours: — Red, RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.
	Title/Meaning/Referent: Acute toxicity Function/description: To indicate the risk of acute toxicity. Level of importance: 1 Colours: — Red RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.

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Table B.8 (continued)

Pictogram	Designation and remarks
	Title/Meaning/Referent: Environmental hazard Function/description: To indicate the risk of environmental hazard. Level of importance: 1 Colours: — Red RGB: 255,0,0 — Black — White Reference: GHS To be used in/on: — Rescue sheet illustration; — Rescue sheet secondary pages under headings 5, 6, 8; — ERG under headings 5, 6, 8, 9.

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- [2] ISO 7202, *Fire protection — Fire extinguishing media — Powder*
- [3] IEC 80416-1, *Basic principles for graphical symbols for use on equipment — Part 1: Creation of graphical symbols for registration*
- [4] UN ECE, *Consolidated Resolution on the Construction of Vehicles (RE3), revision 3*
- [5] UN ECE Regulation No. 110 *Compressed and Liquefied Natural Gas System Components*
- [6] UN ECE GHS REGULATION, Globally Harmonized System of Classification and Labelling of Chemicals (GHS): <http://www.unece.org/trans/danger/publi/ghs/pictograms.html>

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